

AUTOMATIC ROOT CAUSE ANALYSIS BASED ON TRACES FOR LTE SELF-ORGANIZING NETWORKS

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ABSTRACT

Within the functionality of self-healing in self-organizing networks, automatic root cause analysis is typically focused on identifying problems at the cell level, based on the statistics gathered by the operation, administration, and maintenance system. Therefore, mobile operators lose visibility of the problems that directly affect users' performance. Conversely, this article presents a complete strategy to systematically identify, on the basis of information at the user level (by means of mobile traces), why user disconnection occurs (i.e., the cause of the connection's release). The proposed automatic root cause analysis is characterized by a top-down model and provides a comprehensive classification of faults when they are caused by radio-related problems. First, it classifies the connections according to the type of release, and subsequently, it determines the specific fault cause based on the event information for those connections abnormally released. The proposed method for identification of the radio-related cause has been applied in both an LTE simulator and a real LTE network, illustrating the usefulness of the approach.

INTRODUCTION

The greatest concern of mobile operators is to ensure that their customers are provided with the optimal quality of service for the required time. Therefore, one of their priority objectives is to solve the problems in time in order to avoid unintended service disconnections. However, with the advent of the new Long Term Evolution (LTE) networks and the proliferation of a highly varied range of services, the management tasks are becoming increasingly complex. To address this, operators are now putting increasing resources into the automation of the troubleshooting and maintenance tasks through the self-healing (SH) functionality within self-organising networks (SONs) [1]. Briefly, an SH system is responsible for detecting problems in the mobile network, identifying their cause (i.e., root cause analysis, also called diagnosis) and carrying out the compensation and recovery actions [2].

SH has traditionally focused its attention on root cause analysis based on cell-level indicators (i.e., statistics from the operation, administra-

tion, and maintenance system, OAM). Examples include [3, 4], which presented automatic diagnosis systems based on cell-level indicators. The main drawback of that approach is that cell-level indicators represent aggregated and statistical values, which do not provide user-level details, and thus prevent the operator from being able to directly know the actual user-level performance and the cause of unintended service disconnections, which is the main reason users usually complain and, occasionally, lead them to change service provider.

Traditionally, user-level analyses have been performed by means of manual drive tests. However, this is a time-consuming task, since experts have to personally go along a predefined route in order to manually collect the user-level measurements. Thus, the analysis focuses exclusively on those areas sampled during the drive test, excluding analysis of the rest of the areas. Furthermore, manual drive test does not allow operators to analyze the behavior of the connection of each specific subscriber.

Given this situation, the Third Generation Partnership Project (3GPP) has standardized the automatic collection of user-level information by means of mobile traces and the minimization of drive test (MDT) [5]. In particular, traces and MDT use cases are mainly focused on troubleshooting and on analyzing the LTE radio access network (RAN) performance of a particular cell [6, 7]. More specifically, they allow experts to check the radio coverage, test new features, and fine-tune or optimize their network parameters. In addition, trace use cases also include the analysis of user equipment (UE) when it does not work correctly or the identification of network-wide issues that make a specific subscriber suffer some service degradation [6].

However, MDT only automates trace collection, whereas posterior analysis of the traces for troubleshooting purposes is still a manual process. As a result, troubleshooting experts have to invest a lot of time in analyzing the mobile traces in order to discover the problems that each user has suffered during its connection. A thorough analysis focused on user-level data would allow operators to identify the reason for the release, which may be weak coverage, cell unavailability, authentication failure, and so on. However,

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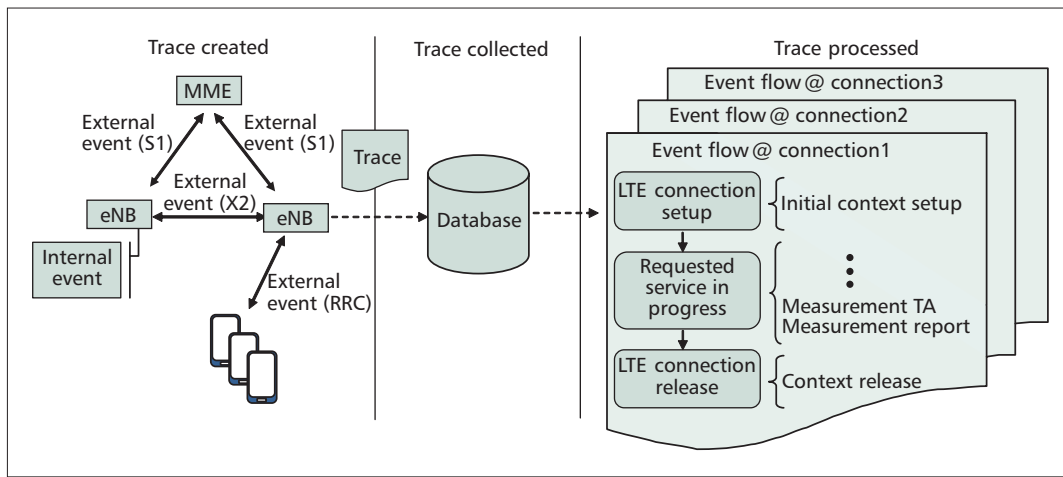


Figure 1. Procedure for trace and event flow collection.

mobile traces involve a huge amount of data, the manual analysis of which is extremely tedious and time-consuming. Consequently, this article proposes to automate mobile traces analysis. For that purpose, big data techniques and current computing power and storage capacity are suitable, since they allow, almost in real time, all this information to be processed. Because of this recent technological development, this topic has not been very relevant until today, but its potential is enormous and establishes the basic principles concerning the SH systems of the future.

Regarding the literature on SH, recent studies show the use of MDT to detect abnormally behaving cells in general [8] and to detect outage cells in particular [9]. Even though the analysis of outage is the most important problem, due to the great impact that it has on the network, there are more types of problems that are worth analyzing. Thus, unlike previous references, this article presents a method to not only detect abnormal behaviors, but also automatically identify a significant group of RF issues, defining the specific cause-indicator relation required to properly automate the root cause analysis.

In this context, this article proposes a unified framework to diagnose the reasons for each user disconnection. This allows automatic discernment between problematic and non-problematic connections, and then focuses attention on the particular connection's stage where the issue lies. Normally, when an undesirable release is caused by problems related to either software or protocols, the events are sufficiently explicit and clear on the cause. However, when there is a radio link failure and thus the release happens due to bad RF conditions, further diagnosis is required. Therefore, the proposed root cause analysis provides a technique to automatically identify the RF problem on the basis of the key indicators reported by the user just moments before the release happened. Then the proposed methodology aims to fully automate the diagnosis of those radio link failures, determining their reason, which is not reported by the RLF event. Furthermore, this approach goes one step further than the RLF event use cases [10], identifying more variety of RF issues (not only coverage holes or mobility problems as proposed in [10]) and automating the whole diagnosis process.

LTE TRACES AND EVENTS

The LTE trace function is used to store radio measurements and signaling messages (i.e., events) in order to reflect user-level performance. Therefore, traces do not aggregate any information but instead record the user-specific values of each event (signal quality, serving cell name, etc.). An event is a report consisting of data related to current system usage and documents all pertinent data regarding the process in question, such as status information (failure or success), a rough idea of the cause of a specific failure (e.g., cell unavailability), the instant when it happened, and even specific measurements and performance information (e.g., signal level, bit rate, timing advance). Those events can be grouped into two categories according to its origin (Fig. 1).

Internal Events: These events represent the performance of the base station (eNB) in the LTE system, so they are specific to each eNB's vendor. Overall, internal events contain information related to all processes that take place at the cell or UE level and user measurements.

External Events: These events are external to the eNB and consist of the signaling messages that the eNB exchanges with the rest of the network equipment. Namely, external events are radio resource control (RRC) messages received from the UE through the LTE-Uu interface and protocol messages that the eNB exchanges, other eNBs through the X2 interface, or the mobility management entity (MME) through the S1 interface. As a consequence, external events can be divided into three categories depending on the type of message they report.

Among all types of events, only those chosen and activated by the operator are registered in a trace file. Operators need to find a balance between the amount of events they need to store in order to ensure accurate analysis and the cost of processing them. Note that the greater the number of activated events and traced cells, the greater the processing time. In particular, at each report period (e.g., 15 min), the eNB provides a binary file containing all received events (Fig. 1). Within these traces, two types can be distinguished depending on the scope of application [6].

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The event flow can be described in three different steps: first, the connection is established; second, the requested service is maintained and provisioned; and finally, the connection is released. In each of these stages, different protocols and network equipment are involved; thus, identifying the phase in which the release occurs provides valuable information on what has happened.

Cell Traffic Trace: Used to record all LTE activated events of all users (or a fraction of them) in the desired cells. Thus, they provide enough information to identify the existing problems in those cells and to analyze in detail the quality of the radio environment.

UE Trace: Only stores the events and the measurements of a specific user selected by the operator. Therefore, with this functionality, network operators can choose the user to be traced using the UE's international mobile subscriber identity (IMSI) or international mobile equipment identity (IMEI).

Given that the scope of this study is to analyze the termination of each user, both types of traces may be useful.

ROOT CAUSE ANALYSIS AT THE UE LEVEL

To identify the reason for a release and to perform a detailed diagnosis when that release is undesired, the end of the user's connection should be analyzed. All events belonging to the same connection are aggregated and temporarily stored, building its event flow (Fig. 1). In general, the event flow can be described in three different steps: first, the connection is established; second, the requested service is maintained and provisioned; and finally, the connection is released. In each of these stages, different protocols (e.g., RRC) and network equipment (e.g., MME) are involved; thus, identifying the phase in which the release occurs provides valuable information on what has happened.

Based on the event flow, UE releases can be grouped into three categories.

Normal Release: A normal release encompasses all releases that happen when the LTE service offered to the user has been completed. The event flow of a UE whose connection ends successfully has a *Context Release* event indicating that the release has been normal. There are different situations where the finalization of the LTE service is considered satisfactory. On one hand, a normal release occurs when no data is transmitted between the LTE network and the end user because either the user has been inactive during a long period (typically more than 10 s) or its session has finished. On the other hand, a normal release can also be due to LTE deployment constraints. For instance, LTE networks that do not support voice call yet have to redirect users who request a voice call to one of the existing 2G/3G networks through the technology known as *CS fallback*.

Access Failure: It takes place in the connection setup phase, which implies that the user cannot obtain the requested service, so its event flow ends with an *Initial Context Setup* event with information about the cause of the failure [11]. An access failure can occur due to several causes, including overload, no radio resources being available, no cell being available, authentication failure, and so on.

Dropped Connection: These are abnormal releases that have negative impact on users because they occur while the requested service is in progress, so there are still buffered data to be transmitted at the time of the release. This kind of release can occur due to hardware errors, breakdown of the interfaces, and failures in any

of the processes of the mobile network (e.g., handovers). All these releases can be diagnosed based on the values of the *Context Release* event [11, 12]. For instance, the connection can be released because of lack of available resources in the target cell during a handover or because a particular timer has expired. However, when the connection drops due to the quality of the air interface, the release cause in the *Context Release* event does not provide such detailed information. In this situation, the cause indicated by the *Context Release* event could be something generic, for example, *Eutran Generated Reason* or *Radio Connection With UE Lost*, and it may be accompanied by a *Radio Link Failure* event [10]. Thus, in order to explain the RF cause of these releases (coverage hole, LTE interference, etc.), further analysis based on user measurements has to be performed.

AUTOMATIC DIAGNOSIS OF RF CAUSES

When an event flow shows that the release is due to bad conditions in the air interface, a well founded analysis has to be carried out in order to reach its radio cause. This section presents the required knowledge to automatically diagnose the radio cause of the dropped connection by means of any classification system: first, an overview of the diagnosis methods proposed in literature is presented; then the required indicators and the characteristics of each radio cause are detailed.

METHODS TO AUTOMATE THE DIAGNOSIS

The automatic diagnosis of the radio cause can be addressed by using any classification system available in the literature. The difference is that instead of applying the classifier to the statistics from the OAM, as proposed in the literature, the inputs to the classifier will be the mobile traces. In particular, the aim of this diagnosis system is to identify the radio cause for each drop based on a set of standard indicators and measurements provided by the event flow.

There are several diagnosis systems available in the literature. On one hand, a rule-based system is the simplest technique to automatically identify the radio cause. This system can easily be designed from the relation between the radio causes and indicators (e.g., Table 1), and it does not require big computational capacity. A more advanced automatic diagnosis system based on a probabilistic method such as Bayesian networks is proposed in [13] for SH in mobile networks. Both systems require the design of thresholds to analyze the input data. There are different ways of doing that, such as the percentile-based discretization (PBD) method proposed in [3]. This method consists on designing the threshold of each key performance indicator (KPI) based on their X th percentile defined by the experts. That is, each threshold will take the X th percentile of the KPI values in the training dataset. For instance, if the expert chooses percentage 90 for the reference signal received quality (RSRQ), its associated threshold is set to the 90th percentile of all the values of RSRQ in the dataset.

On the other hand, the root cause analysis of radio problems can also be automated using either the ranking method proposed in [4] or a

Radio cause	Serving RSRP	Serving RSRQ	Strongest RSRP	Number of cells	Relative TA
CH	$<Thr_{RSRP_B}$	$<Thr_{RSRQ}$	$<Thr_{RSRP_B}$	$<Thr_{NC}$	$<Thr_{TA}$
CE	$<Thr_{RSRP_B}$	$<Thr_{RSRQ}$	-	-	$\geq Thr_{TA}$
LD	$<Thr_{RSRP_B}$	$<Thr_{RSRQ}$	$<Thr_{RSRP_B}$	$\geq Thr_{NC}$	$<Thr_{TA}$
MP	$<Thr_{RSRP_G}$	$<Thr_{RSRQ}$	Better than serving	-	$<Thr_{TA}$
I	$\geq Thr_{RSRP_G}$	$<Thr_{RSRQ}$	-	-	$<Thr_{TA}$

Table 1. Radio cause and indicator relation.

neural network such as the self-organizing map [14], which do not require any threshold to analyze the input data. Unlike the rule-based system, these methods are more complex and require more computational cost, but if they are properly designed on the basis of a good set of real samples or solved cases, the diagnosis success rate will be improved.

Thus, among the available methods, the operator should select the most appropriate technique for its priorities, requirements, and available resources.

INDICATORS AND MEASUREMENTS

The proposed indicators are those that provide enough information to analyze the RF conditions of each user when its connection is lost. This information is taken from the required events that have happened just before the end of the event flow. For the proposed framework, only two events are required: the *Measurement Report*, which provides the reference signal received power (RSRP) and the RSRQ of all the cells measured by the user; and the *Measurement TA*, which determines the value of the timing advance (TA) [15], which represents the transmission delay in the downlink and uplink path between the user and its serving cell. Both the user measurement and its location can also be obtained from the *Radio Link Failure* event [10]. Note that this event is only available when the user has experienced an RLF, so if the values reported by normal users are required in order to compare the values or design thresholds, they should be obtained from the *Measurement Report* and *Measurement TA* events. Furthermore, it is worth mentioning that the availability and accuracy of the location information depend on the used location estimation method and LTE release. For example, a precise location can be obtained when the Global Positioning System (GPS) system is available; otherwise, the position can be estimated from the TA. The proposed indicators are:

- **Serving RSRP:** It is the signal strength that the UE measures from the serving cell [16] when its connection finishes. A decrease of the RSRP can be caused by either an increase of the distance or obstacles on the path. Therefore, to distinguish between those situations, this indicator should be complemented with information about the separation between the UE and the site.

- **Serving RSRQ:** The quality of the received

reference signal from the serving cell is crucial to understanding the cause of the release. RSRQ is defined in [16] as the ratio of total reference power to the total received power within the measurement bandwidth.

- **Strongest non-serving RSRP:** The RSRP of the strongest cell (other than the serving one) is also important. It represents the best non-serving cell measured by the UE, thus providing additional information regarding the global LTE coverage.

- **Number of detected cells:** The number of non-serving cells that the UE has measured at the time of the release indicates the number of cells that might cause interference to the user's connection. Thus, this indicator allows areas with large cell overlaps to be distinguished.

- **Relative TA:** The radio cause also depends on the location where the release has taken place. Therefore, the TA is used to determine the distance between the UE and its serving cell. However, depending on the antenna configuration, height, and location, the service area will be different. This implies that users with the same TA but belonging to different cells could be considered to be near or far according to the service area. This means that the relative TA of the UE with respect to the service area of its cell should be considered. This requires identifying the real distance within which the majority of the users are served by a particular cell, so two regions can be differentiated: the cell center, where the UEs are considered to be near the cell, and the cell edge, where the farthest users are located. The border between those regions can be estimated as the Xth percentile (e.g., 98th) of all TA values reported in that cell. As a result, the relative TA is the ratio of the user TA to the estimated Xth percentile.

RADIO CAUSES

When analyzing the RF conditions of the UE at the time of the release, different radio causes can be found related to both coverage and interference. The specific features of those radio causes along with the expected behavior of the indicators are detailed below and summarized in Table 1.

- **Coverage Hole (CH):** It is an area where both the serving and the strongest RSRPs are insufficient to provide and maintain the LTE service with the quality requirements; specifically, they are below the threshold under which values of

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Parameter	Simulated network	Live network
Grid	Urban area	Urban area
Number of sites	25	25
Number of macrocells	75	75
System bandwidth	1.4 MHz	10 MHz
Max. transmit power	43 dBm	46 dBm
Antenna downtilt	6°	4° (on average)
Handover margin	2 dB	2 dB
Trace reported period	1 simulation loop (18,000 simulation steps)	15 min
Observation period	4 simulation loops	11:00–14:00
Days under observation	–	4
Thr_{RSRPG}	– 84.8 dBm	– 86 dBm
Thr_{RSRPB}	– 107.9 dBm	– 111 dBm
Thr_{RSRQ}	– 13.1 dB	– 7.5 dB
Thr_{NC}	3	3
Thr_{TA}	1	1

Table 2. Parameters of LTE networks.

RSRP are considered bad (Thr_{RSRPB}). This lack of signal power in a zone of the coverage area can be caused by inadequate antenna parameters, wrong RF planning, or physical obstacles. Furthermore, the quality of the serving cell is beneath its corresponding threshold (Thr_{RSRQ}). An area with these RF conditions (low RSRP, low RSRQ, and low strongest RSRP) is considered to be a coverage hole provided that it is located within the coverage area. Thus, the operators should define the threshold Thr_{TA} to determine the border between cell edge and cell center. The lower the Thr_{TA} , the lower the percentage of serving area is considered as a cell center. In addition, since a CH is characterized by low overlap of cells, the number of cells detected by the UE should be lower than the specific threshold (Thr_{NC}).

Lack of Dominant Cells (LD): This problem takes place in an interference-limited environment within the cell center of the serving cell, where there are no dominant cells, including the server. Consequently, the user's relative TA is lower than the TA threshold, and the excessive overlap between LTE cells can be identified when the UE measures a high number of LTE cells (higher than the threshold). Nevertheless, none of them provides a clear and powerful signal for maintaining the connection (the RSRP of both serving and strongest cells and the serving RSRQ are below the thresholds), and thus the service fails. In some situations, CH and LD can be very similar; however, they can be dis-

tinguished by either the high overlap (LD) vs. regions that are covered by a few cells (CH). The number of measured cells determining high overlap (Thr_{NC}) is decided by the operator on the basis of the network requirements.

Cell Edge (CE): The serving RSRP and RSRQ are below the specified thresholds mainly because of the high propagation loss caused by excessive distance between the base station and the UE. Therefore, the relative TA is the key indicator to distinguish the CE from the rest of these situations. In particular, it should be above the considered threshold. It can reasonably be expected that users at the CE measure other cells, so the value of the strongest RSRP and the number of measured cells are not considered relevant.

Mobility Problems (MP): The release is caused by MP when the user may not have sufficient signal strength or quality to maintain the session (i.e., serving RSRP below Thr_{RSRPG} and RSRQ below Thr_{RSRQ}). However, there is another cell that could have continued its session avoiding the abnormal release (i.e., the strongest RSRP is better than the serving one), but the user does not carry out the handover to the strongest cell and the connection drops. In this situation the value of relative TA should be small (i.e., below its threshold) in order to ensure that the user is not very far in relation to the coverage area of its serving cell (where the propagation delay could be too high to allow a successful communication with the serving cell to perform a handover). Unlike the other problems, the number of cells is not necessary.

Interference (I): A user suffering from I has an RSRP above the threshold, but its connection is lost due to bad quality of the air interface. Therefore, the serving RSRQ is lower than the corresponding threshold. The number of cells measured by the user could differ considerably, so this indicator is insubstantial for this radio cause.

CASE STUDY: DIAGNOSING USER DISCONNECTIONS

Once the theoretical basis of the framework has been presented, in this section, the method is evaluated using both simulated and real data. In particular, the aim of this feasibility study is to illustrate how the proposed framework works, showing the usefulness and potential of automating the diagnosis of user disconnections using mobile traces.

The rule-based system has been chosen among all automatic classification systems proposed in the literature due to its simplicity and similarity with the human thought process. It is characterized by being analogous to the current manual procedure carried out by experts, and thus it provides the diagnosis in a similar manner to human thinking. As a result, the rule-based system is especially suited for demonstrating the proposed framework, providing a straightforward evaluation and easy-to-understand results.

To automate the process with a rule-based system, the criterion of each radio cause is translated in terms of IF...THEN rule. In particular, each rule is easily designed on the basis of the

relation between the radio causes and indicators (presented in Table 1). For instance, the rule of CH would be like this: “**IF** ($RSRP_{Serving} < Thr_{RSRP_B}$) **AND** ($RSRQ_{Serving} < Thr_{RSRQ}$) **AND** ($RSRP_{Non-Serving} < Thr_{RSRP_B}$) **AND** ($Num_{Cell} < Thr_{NC}$) **AND** ($TA_{Rel} < Thr_{TA}$), **THEN** (Radio Cause is Coverage Hole).”

Finally, the required thresholds for the RSRP and RSRQ indicators have been set automatically through the percentile-based discretization (PBD) method [3]. However, the thresholds of the other indicators (Num_{Cell} and TA_{Rel}) have been set according to the operator’s strategy, which defines the acceptable level of overlap (i.e., the number of detectable cells) and the adequate distance to the cell border (e.g., $Thr_{NC} = 3$ and $Thr_{TA} = 1$, in this study).

SIMULATIONS

Simulated Network: In order to prove the automatic root cause analysis, a real LTE network from an urban area has been simulated in MATLAB with the dynamic system-level LTE simulator presented in [17] and with the general simulation setup used in [14]. Briefly, the propagation model used is Okumura-Hata with wrap-around and log-normal slow fading. Furthermore, the specific simulation parameters are presented in Table 2. In particular, the considered part of the real network consists of 25 tri-sectorial macrocells. Among all of them, five different cells have been configured to have a specific RF problem, which facilitates the analysis of each of them individually. Each proposed radio cause has been modeled as follows (Fig. 3):

- CH: generated within the coverage area of $Cell_{1B}$ by increasing the propagation loss in a specific small square zone
- LD: modeled by changing the antenna tilt of cells $Cell_{5A}$, $Cell_{6A}$, and $Cell_{1B}$ so that there cannot be any dominant server in the intersection among their coverage areas
- CE: modeled by significantly increasing the downtilt of $Cell_{1A}$ so that its coverage area is reduced, causing an expansion of the cell edge zone
- I: caused by an external antenna within the coverage area of $Cell_{2A}$
- Mobility problems: caused by misconfiguring the mobility parameters between $Cell_{4A}$ and $Cell_{3A}$

Experimental Results: The proposed method has been used to identify the cause of each abnormal release that took place in those problematic cells. It is important to note that the design of thresholds through the PBD method depends on the specified X th percentile and the characteristics of the dataset. In addition, since the performance of the rule-based system depends on the designed thresholds, a sensitivity analysis has been performed. In particular, Fig. 2 represents the receiver operating characteristic (ROC) of the diagnosis system for each radio cause individually, using different thresholds. In general, the overall performance does not vary significantly as the thresholds are varied, since the dataset is mainly composed of normal cases. By comparing the results of the diagnosis for each radio cause with the best ones, it can be appreciated that the false positive rate slightly increases when any of

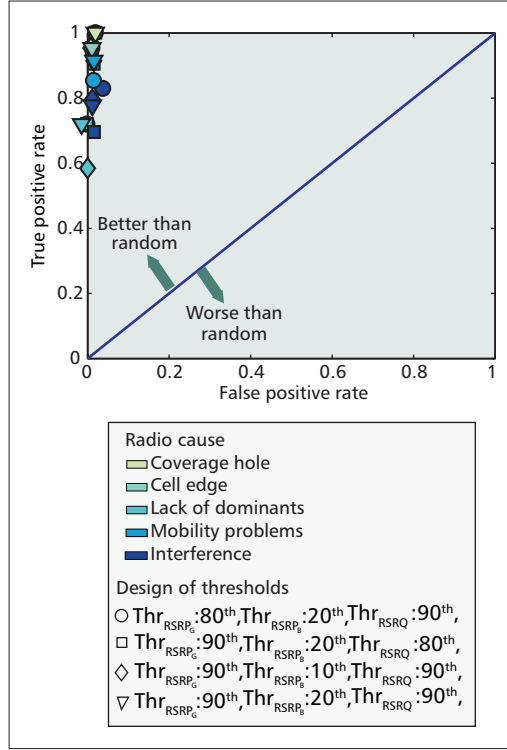


Figure 2. ROC plot of the four diagnosis systems where each color represents the radio cause, and each symbol determines the configuration of the diagnosis system.

the thresholds becomes less strict (e.g., I when $Thr_{RSRP_G} = 80$ th). Conversely, as the thresholds are more strict, the activation of the rule is more difficult, so the true positive rate decreases (e.g., LD when $Thr_{RSRP_B} = 10$ th). Hence, the best results are achieved with $Thr_{RSRP_G} = 90$ th, $Thr_{RSRP_B} = 20$ th, and $Thr_{RSRQ} = 90$ th, since it reaches a compromise between false positive rate and true positive rate. Based on this evaluation, the rest of the analysis has been performed with that configuration of thresholds; their values are shown in Table 2.

In Fig. 3a the disconnections that happened in $Cell_{1A}$, $Cell_{1B}$, $Cell_{2A}$, $Cell_{4A}$, and $Cell_{5A}$ have been represented over the LTE network in the location where their connection was lost. Each dot represents an abnormal release, and they are plotted with a specific color depending on their diagnosed radio cause. Note that those disconnections that do not fulfill any rule have been labeled as not identified (NI) in Fig. 3. Furthermore, Fig. 3c shows, for each analyzed cell, the histogram of the dropped connections given each radio cause.

Users in the Coverage Hole: By carefully looking at the dropped connections of $Cell_{1B}$ (Fig. 3), it is noticeable that there is a high concentration of connections dropped due to the CH.

Users at CE: Other types of users who lose their connections due to bad RF signal and quality correspond to those located at the CE of $Cell_{1A}$, beyond the considered coverage area. This class allows experts to easily distinguish between users that are within the desired coverage area with bad RF conditions and those with RF issues that are caused by CE conditions.

A user suffering from interference has an RSRP above the threshold but its connection is lost due to bad quality of the air interface. Therefore, the serving RSRQ is lower than the corresponding threshold. The number of cells measured by the user could differ considerably, so that this indicator is insubstantial for this radio cause.

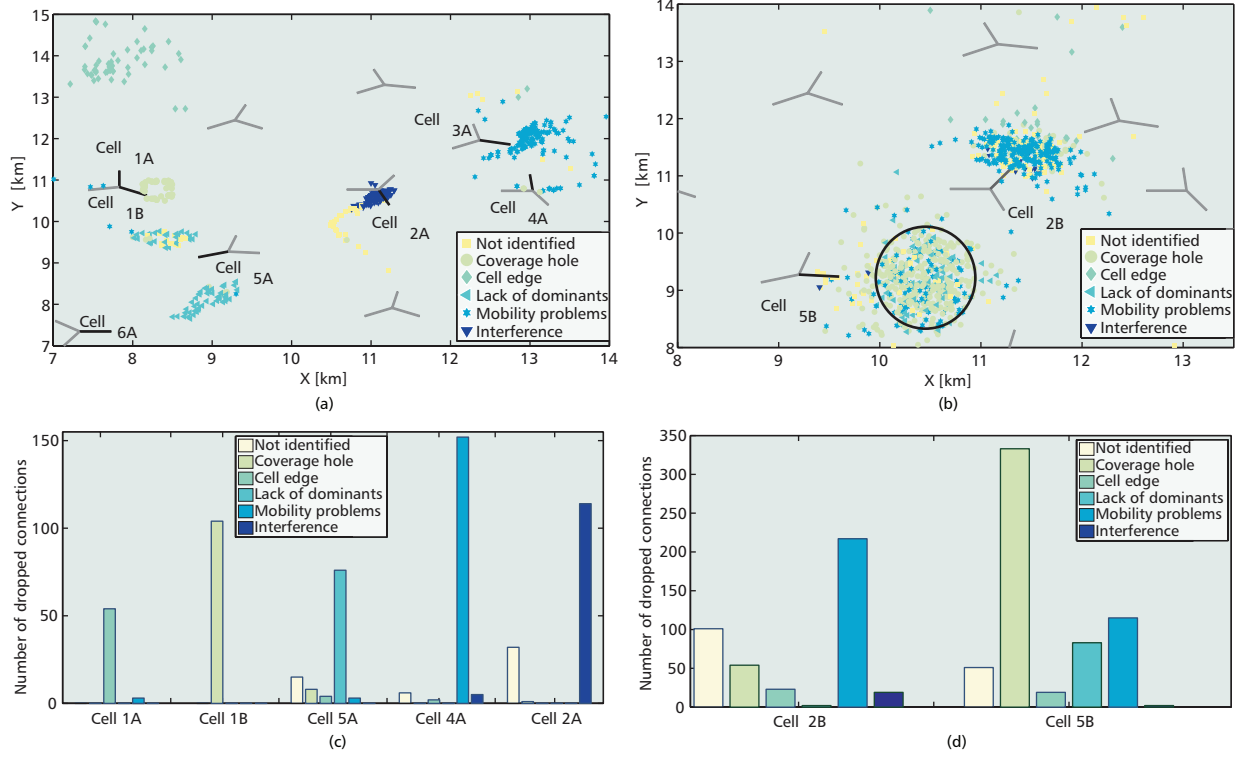


Figure 3. Results of the automatic root cause analysis: a) and c) show the dropped connections in the simulated LTE network, while (b) and (d) present the results of the live network. a) and b) show the location of each dropped connection, while c) and d) show the total number of dropped connections in each cell, grouped by their RF cause.

Users with Lack of Dominant Cells: Figure 3 also reveals that there is a weak coverage area perched at the intersection between *Cell*_{5A} and *Cell*_{6A} and between *Cell*_{5A} and *Cell*_{1B}. Users in that area measure several cells, but none of them provides a good level of signal that leads to a successful connection. As a result, there is a high concentration of disconnection due to LD in those areas. It is appropriate to clarify that abnormal releases of both *Cell*_{5A} and *Cell*_{1B} are shown simultaneously in Fig. 3, so some of them are overlapped.

Users Suffering Interference: In *Cell*_{2A}, there are some users that receive adequate signal level from their serving cell but suffer from poor quality due to the interference of a nearby antenna.

Users with Mobility Problems: In Fig. 3, a lot of connections of *Cell*_{4A}, have finished because of mobility problems. This means that these users received bad signal from their serving cell while receiving better signal from an adjacent cell, and they have lost their connection instead of performing a handover.

The majority of the abnormal releases have been properly diagnosed according to the problem of their cell (Table 3a). In this table the number of RF faults diagnosed in each faulty cell is shown specifying the problem created in those cells. For simplicity, as the reference diagnosis, all abnormal releases in each cell are considered to be caused by the created fault, with the exception of the abnormal releases of *Cell*_{1B}, which are considered to be caused by the CH if and only if they take place in the shadow area. Furthermore, the values of some typical evaluation metrics

[18] are presented in Table 3b, for instance, the average precision is 94.81 percent and average error rate is 3.63 percent. Note that the chosen automatic classification system is the simplest one and uses crisp thresholds, which makes the proper identification more difficult. More sophisticated diagnosis systems, such as those in [4, 13, 14], should improve those results. In spite of this, the obtained error rate is comparable to that obtained by a human expert.

LIVE NETWORK

The proposed method has been applied to the real LTE network of the same urban area. The most relevant characteristics of this network are described in Table 2 along with the experimental setup used in this study. In particular, the data of the live network has been collected each report period (i.e., 15 min) during 4 days from 11:00 to 14:00. Then each disconnection that took place during this period of time has been analyzed. Therefore, the results are detailed values that represent the diagnosis obtained for each specific disconnection.

The analysis of this network has been focused on two problematic cells (*Cell*_{2B} and *Cell*_{5B}). A conventional non-automated analysis performed by operators identified a weak coverage area, indicated by a circle in Fig. 3b, and a cell (*Cell*_{2B}) with lots of drops due to mobility problems, which reduced its handover success rate.

Results obtained with the proposed method are shown in Fig. 3, representing the location of the abnormal releases of both cells due to RF problems in Fig. 3b, and the histogram with the

			(a)						(b)	
			Diagnosis						Performance measure [%]	
			CH	CE	LD	MP	I	NI		
Created problem	Cell _{1B}	CH	104	0	0	0	0	0	Average accuracy	96.37
	Cell _{1A}	CE	0	54	0	3	0	0	Average precision	94.81
	Cell _{5A}	LD	8	4	76	3	0	15	Average recall	87.22
	Cell _{4A}	MP	0	2	0	152	5	6	Average F-score	90.86
	Cell _{2A}	I	1	0	0	0	114	32	Average error rate	3.63

Table 3. Experiment results.

number of dropped connections of each type in Fig. 3d. The majority of the dropped connections in *Cell_{5B}* are concentrated in the area identified by the operator as weak coverage, and they have been automatically diagnosed by the proposed method as users in a CH. This means that those users do not receive the good levels of LTE signal that lead to a successful connection, from neither its serving cell nor a neighboring cell. As a result, it is reasonable to suggest that the radio cause diagnosed by the proposed system in a shadow area is consistent with the information provided by the operator. Figure 3b also reveals that within the weak coverage area, there are some users that have been identified as users without a dominant server. This is also a consequence of this problem given that it has caused some users to eventually measure other neighboring cells, but none is capable of providing service.

Furthermore, results in Fig. 3 also show that *Cell_{2B}* presents a high concentration of abnormal releases due to mobility problems within its coverage area. This is in line with the analysis carried out by operators, which revealed that the performance of this cell was deteriorated due to handover issues. In addition, *Cell_{2B}* has some dropped connections due to CE conditions that correspond with the furthest users.

CONCLUSIONS

An automatic method for root cause analysis of user disconnections has been presented. In particular, this article details an approach based on traces to identify the cause of the release, with a particular emphasis on abnormal release due to RF issues. In addition, the main challenges of using data traces and performing the diagnosis at the user level are addressed. Through the use of mobile traces, the automatic diagnosis can be performed almost in real time, providing plenty of detail on the problems in the mobile network. An additional benefit is that the proposed strategy only requires a few standard indicators to determine the RF problem. Therefore, with this system the automatic diagnosis of self-organizing networks is substantially improved.

The RF classification of the proposed technique has been assessed through a rule-based system in a real and simulated LTE network. The obtained results show the advantages of knowing

the specific radio causes of the problems that adversely affect each user in each location and how the proposed root cause analysis provides sensible and coherent classification.

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Note that the chosen automatic classification system is the simplest one and it uses crisp thresholds, which makes the proper identification more difficult. More sophisticated diagnosis systems should improve those results. In spite of this, the obtained error rate is comparable to that obtained by a human expert.

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BIOGRAPHIES

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